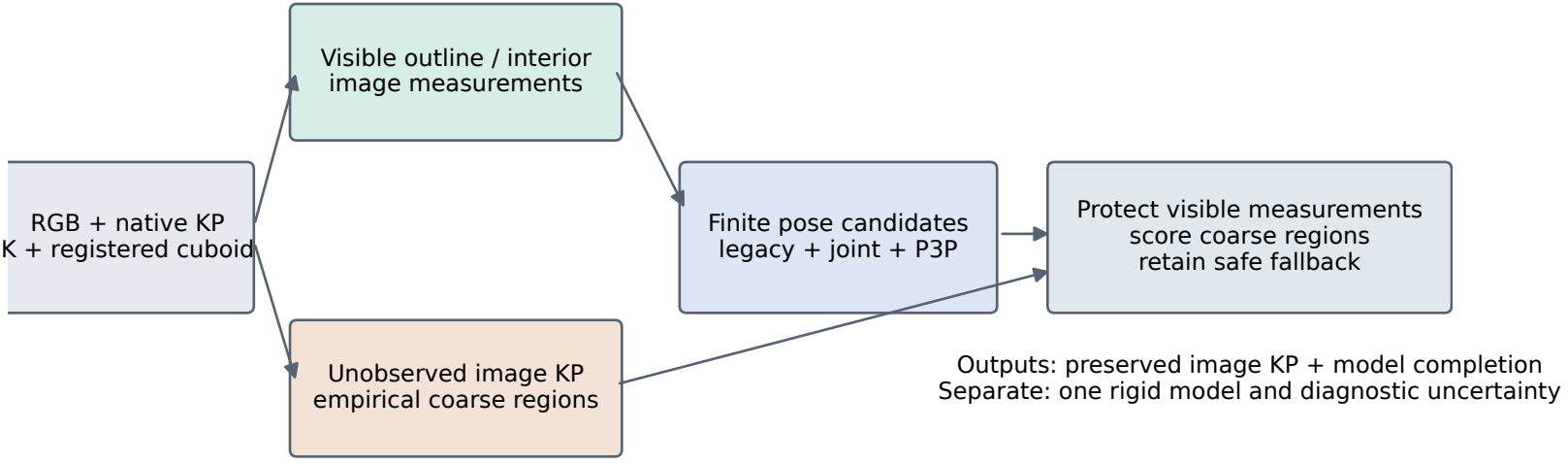
RGB + native KP
K + registered cuboid
Visible outline / interior
image measurements
Unobserved image KP
empirical coarse regions
Finite pose candidates
legacy + joint + P3P
Protect visible measurements
score coarse regions
retain safe fallback
Outputs: preserved image KP + model completion
Separate: one rigid model and diagnostic uncertainty